

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] DECEMBER 2024

Paper Code: ES-201

Subject: Computational Methods

Time: 3 Hours

Maximum Marks: 60

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit. Scientific calculator is allowed.

Q. No.	Question (Q1 is Compulsory)	Marks														
1(a)	Determine the decimal number that corresponds to the machine word $[45DE4000]_{16}$.	[2]														
1(b)	Using secant method find the root of $x \sin(x) - 3 \cos(x) = 0$ between $(0.0, 1.8)$ with accuracy of 2 digits after decimal point.	[3]														
1(c)	For following given set of data for x & y, formulate the Newton divided differences table:	[3]														
	<table><tr><td>x</td><td>0.00</td><td>2.00</td><td>4.00</td><td>6.00</td><td>8.00</td></tr><tr><td>y</td><td>0.00</td><td>8.00</td><td>64.00</td><td>216.00</td><td>512.00</td></tr></table>		x	0.00	2.00	4.00	6.00	8.00	y	0.00	8.00	64.00	216.00	512.00		
	x		0.00	2.00	4.00	6.00	8.00									
y	0.00	8.00	64.00	216.00	512.00											
1(d)	Compute $\int_0^1 [\sin(x) / x] dx$ by using composite Trapezoid rule with six uniform points. Assign the value $\sin(x)/x = 1$ at $x = 0$.	[3]														
1(e)	Define Decomposition of any matrix. Explain the required condition for Doolittle and Crout decomposition method.	[2]														
1(f)	Find the value of λ for which the system of equations $x + y + 4z = 1$, $x + 2y - 2z = 1$ & $\lambda x + y + z = 1$ will have unique solution.	[3]														
1(g)	Using Euler's method find the value of y at $x = 0.10$ of the ODE $dy/dx = x + y + xy$ where initial conditions are $x_0 = 0.00$ & $y_0 = 1.00$ and using step size 0.10.	[3]														
1(h)	Give an example and conditions of Parabolic, Hyperbolic and Elliptic partial differential equations.	[1]														
UNIT-I																
2(a)	Find the value of function $f(x) = 1 - \cos(x)$ at $x = 0.1$. Modify the expression so that loss of significant digits can be avoided and calculate the value again. Compare two values with the true value 0.4996×10^{-2} .	[5]														
2(b)	Define rate of convergence and stability of iterative method. Prove that the rate of convergence of Newton-Raphson method is 2.	[5]														
3(a)	[OR] Define Multivariate unconstrained minimization problem with an example. Using Newton Method to minimize multivariate function, minimize $f(x) = 4x_1^2 + x_2^2 - 2x_1x_2$ starting at $x^{(0)} = [1, 1]^T$.	[5]														
3(b)	Determine the minimum point of the function $f(x) = x^2 - 7x + 12$ by Fibonacci search method, if the first uncertainty interval is [2, 4].	[5]														
UNIT-II																
4(a)	Explain error estimation in Newton-Gregory Forward interpolation. Also prove that the maximum error in Newton's Forward interpolation is 1 when $ x - x_0 < h$, where h is step size of the given data.	[5]														
4(b)	For the given set of data for X and Y construct table of forward differences. Find interpolated value of Y at $X = 4.60$ using Newton forward interpolation formula. Also find estimated error:	[5]														
	<table><tr><td>X</td><td>1.0</td><td>2.0</td><td>3.0</td><td>4.0</td><td>5.0</td><td>6.0</td></tr><tr><td>Y</td><td>7.0</td><td>13.0</td><td>21.0</td><td>32.0</td><td>48.0</td><td>70.0</td></tr></table>		X	1.0	2.0	3.0	4.0	5.0	6.0	Y	7.0	13.0	21.0	32.0	48.0	70.0
	X		1.0	2.0	3.0	4.0	5.0	6.0								
Y	7.0	13.0	21.0	32.0	48.0	70.0										

[P.T.O.]

Q. No.	Question (Continued)	Marks																									
5(a)	[OR] Find $I = \int_{-1}^1 [1 / (1 + x^2)] dx$ by Gaussian Quadrature formula for $n = 4$, where values of Abscissae = (± 0.33998 & ± 0.86114) and Weights = (0.65214 & 0.34785).	[4]																									
5(b)	Evaluate definite integral $\int_0^1 [1 / (1 + x^2)] dx$ correct to three decimal places using basic trapezoidal rule with $h = 0.5, 0.25$ and 0.125 then obtain improved estimate using Romberg's method. Compare results with true value.	[6]																									
UNIT-III																											
6(a)	Test consistency of system of equations: $5x_1 + 3x_2 + 7x_3 = 4, 3x_1 + 26x_2 - 2x_3 = 9, 7x_1 + 2x_2 + 10x_3 = 5$. Does the system have finite or infinite solutions?	[4]																									
6(b)	Explain partial pivoting and solve by Gauss-Jordan method: $x_1 + x_2 + x_3 = 1, 4x_1 + 3x_2 - x_3 = 6, 3x_1 + 5x_2 + 3x_3 = 4$	[6]																									
7(a)	[OR] Obtain linear Spline for function $f(x)$ defined by data below, and evaluate at $x = 3$: <table><tr><td>x</td><td>1</td><td>2</td><td>4</td></tr><tr><td>$f(x)$</td><td>3</td><td>7</td><td>21</td></tr></table>	x	1	2	4	$f(x)$	3	7	21	[4]																	
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$f(x)$	3	7	21																								
7(b)																											
UNIT-IV																											
8(a)	Using Picard method, compute $y(0.2)$ to three decimal precision from $dy/dx = x + y$ given $y(0) = 1$. Compare result with exact result for value of y at 0.2.	[4]																									
8(b)	Consider initial value problem $dy/dx = x + y^2$ where $y(0) = 1$. Using Runge-Kutta method of 4th order find $y(0.2)$ taking $h = 0.1$.	[6]																									
9(a)	[OR] Solve initial value problem $dy/dx = x - y^2$ in range $0 < x < 1$, for initial condition $y(0) = 0$.	[4]																									
9(b)	Solve elliptical equation $u_{xx} + u_{yy} = 0$ for the following square mesh with boundary values given as: <table><tr><td>0</td><td>500</td><td>1000</td><td>500</td><td>0</td></tr><tr><td>1000</td><td></td><td></td><td></td><td>1000</td></tr><tr><td>1000</td><td></td><td></td><td></td><td>1000</td></tr><tr><td>1000</td><td></td><td></td><td></td><td>1000</td></tr><tr><td></td><td>500</td><td>1000</td><td>500</td><td></td></tr></table>	0	500	1000	500	0	1000				1000	1000				1000	1000				1000		500	1000	500		[6]
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